Year 10 Trigonometry (part 2) - Homework Answers

Revision | Warm-up

Angles are rounded only at the last step. The sign $\fallingdotseq$ means "approximately equal to".

1 Year 11 Advanced Exercise 5B Q4

In ΔXYZ the right angle is at Y. From θ, the side $YZ = 61$ is opposite and $XY = 78$ is adjacent.

$$\tan\theta = \frac{61}{78}$$
$$\theta \fallingdotseq 38.03°$$

The bearing is measured clockwise from north, and XY points north, so the bearing of Z from X is 038°T.

2 Year 11 Advanced Exercise 5B Q7

Let S be the start, E the end of the 120 km leg, and F the point due south of S. The angle between north and SE is 130°, so the angle between south and SE is $\angle FSE = 180° - 130° = 50°$. In the right-angled ΔSFE, SF is adjacent to the 50° angle and $SE = 120$ km is the hypotenuse.

$$\frac{SF}{120} = \cos 50°$$
$$SF = 120\cos 50°$$
$$\fallingdotseq 77.13$$

She is about 77 km from her starting point.

3 Year 11 Advanced Exercise 5B Q8

The horizontal at W is parallel to the street, so $\angle WCB = 73°$ (alternate angles). In ΔWBC, WB is opposite this angle and $BC = 7$ m is adjacent.

$$\frac{WB}{7} = \tan 73°$$
$$WB = 7\tan 73°$$
$$\fallingdotseq 22.90$$

The window is about 23 m above the street.

4 Year 11 Advanced Exercise 5K Q1

a AAS: sine rule. The side x is opposite 57° and the side 23 is opposite 43°.

$\frac{x}{\sin 57°} = \frac{23}{\sin 43°}$, so $x = \frac{23\sin 57°}{\sin 43°} \fallingdotseq 28.3$

b Right-angled: x is opposite 52° and 13.5 is adjacent.

$\frac{x}{13.5} = \tan 52°$, so $x = 13.5\tan 52° \fallingdotseq 17.3$

c SAS: cosine rule, with the included angle 62°.

$x^2 = 11^2 + 13^2 - 2\times 11\times 13\times\cos 62° = 290 - 286\cos 62° \fallingdotseq 155.73$, so

$x \div 12.5$

d Right-angled: 24 is opposite 41°30′ and x is the hypotenuse.

$\frac{x}{24} = \frac{1}{\sin 41°30'}$, so $x = \frac{24}{\sin 41°30'} \div 36.2$

e SAS: cosine rule, with the included angle 98°.

$x^2 = 7.4^2 + 9.2^2 - 2 \times 7.4 \times 9.2 \times \cos 98°$

$= 139.4 - 136.16 \cos 98° \div 158.35$, so $x \div 12.6$

f The third angle is $180° - 54° - 49° = 77°$. The side x is opposite 77° and 18 is opposite 49°.

$\frac{x}{\sin 77°} = \frac{18}{\sin 49°}$, so $x = \frac{18 \sin 77°}{\sin 49°} \div 23.2$

5 Year 11 Advanced Exercise 5K Q2

a SSS: cosine rule. The angle θ is between the sides 10 and 12, and opposite 11.

$\cos\theta = \frac{10^2 + 12^2 - 11^2}{2 \times 10 \times 12} = \frac{123}{240}$, so $\theta \div 59.17° \div 59°$

b Sine rule: θ is opposite 8, and 67° is opposite 9.

$\frac{\sin\theta}{8} = \frac{\sin 67°}{9}$, so $\sin\theta = \frac{8 \sin 67°}{9} \div 0.8182$ and $\theta \div 54.91° \div 55°$.

θ is opposite the side 8, which is shorter than the side 9 opposite 67°, so $\theta < 67°$ and θ cannot be obtuse.

c Right-angled: 13 is adjacent to θ and 17 is the hypotenuse.

$\cos\theta = \frac{13}{17}$, so $\theta \div 40.12° \div 40°$

6 Year 11 Advanced Exercise 5K Q7

a $\angle ABC$ is between $AB = 4$ and $BC = 7$, and opposite $CA = 5$.

$$5^2 = 4^2 + 7^2 - 2 \times 4 \times 7 \times \cos B$$
$$56 \cos B = 16 + 49 - 25 = 40$$
$$\cos B = \frac{40}{56} = \frac{5}{7}$$
$$B \div 44.415°$$

$0.415° \times 60 \div 25'$, so $\angle ABC \div 44°25'$.

b $\text{area} = \frac{1}{2} \times AB \times BC \times \sin B = \frac{1}{2} \times 4 \times 7 \times \sin 44°25'$

$= 14 \sin 44°25' \div 9.80$, so the area is about 10 cm².

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T1 | Angles of any size

Lesson Examples

C1 **Tutor demo.**

a Add or subtract 360°: $50° + 360° = 410°$, $410° + 360° = 770°$, $50° - 360° = -310°$, $-310° - 360° = -670°$. So 410°, 770°, −310°, −670° (other answers differ from these by multiples of 360°).

b i $-120° + 360° =$ 240° **ii** $470° - 360° =$ 110°

Your turn.

a $80° + 360° = 440°$, $440° + 360° = 800°$, $80° - 360° = -280°$, $-280° - 360° = -640°$. So 440°, 800°, −280°, −640°.

b i $-150° + 360° =$ 210° **ii** $520° - 360° =$ 160°

C2 **Tutor demo.**

a $OP = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$, so P is on the circle.

b $x = -5$, $y = 12$, $r = 13$: $\sin\theta = \frac{y}{r} = \frac{12}{13}$, $\cos\theta = \frac{x}{r} = -\frac{5}{13}$, $\tan\theta = \frac{y}{x} = \frac{12}{-5} = -\frac{12}{5}$

Your turn.

a $OP = \sqrt{8^2 + (-15)^2} = \sqrt{64 + 225} = \sqrt{289} = 17$, so P is on the circle.

b $x = 8$, $y = -15$, $r = 17$: $\sin\theta = -\frac{15}{17}$, $\cos\theta = \frac{8}{17}$, $\tan\theta = \frac{-15}{8} = -\frac{15}{8}$

C3 **Tutor demo.**

a $450° - 360° = 90°$, so the point is $(0, r)$: $\sin 450° = \frac{r}{r} =$ 1, $\cos 450° = \frac{0}{r} =$ 0, $\tan 450° = \frac{r}{0}$ is undefined.

b $-90° + 360° = 270°$, so the point is $(0, -r)$: $\sin(-90°) =$ -1, $\cos(-90°) =$ 0, $\tan(-90°)$ is undefined.

Your turn.

a $540° - 360° = 180°$, so the point is $(-r, 0)$: $\sin 540° =$ 0, $\cos 540° =$ -1, $\tan 540° = \frac{0}{-r} =$ 0.

b $-270° + 360° = 90°$, so the point is $(0, r)$: $\sin(-270°) =$ 1, $\cos(-270°) =$ 0, $\tan(-270°)$ is undefined.

C4 **Tutor demo.**

a 115° is between 90° and 180°: quadrant 2 (S). $\sin$ +, cos −, tan −

b 250° is between 180° and 270°: quadrant 3 (T). $\sin$ −, cos −, tan +

c $-60° + 360° = 300°$: quadrant 4 (C). $\sin$ −, cos +, tan −

Your turn.

a 165°: quadrant 2 (S). $\sin$ +, cos −, tan −

b 320°: quadrant 4 (C). $\sin$ −, cos +, tan −

c $400° - 360° = 40°$: quadrant 1 (A). $\sin$ +, cos +, tan +

C5 **Tutor demo.**

a $480° - 360° = 120°$: quadrant 2, related angle $180° - 120° = 60°$, sin positive.

$\sin 480° = +\sin 60° = \dfrac{\sqrt{3}}{2}$

b $-135° + 360° = 225°$: quadrant 3, related angle $225° - 180° = 45°$, cos negative. $\cos(-135°) = -\cos 45° = -\dfrac{1}{\sqrt{2}}$

c $660° - 360° = 300°$: quadrant 4, related angle $360° - 300° = 60°$, tan negative.

$\tan 660° = -\tan 60° = -\sqrt{3}$

Your turn.

a $600° - 360° = 240°$: quadrant 3, related angle 60°, sin negative.

$\sin 600° = -\sin 60° = -\dfrac{\sqrt{3}}{2}$

b $-330° + 360° = 30°$: quadrant 1, related angle 30°, cos positive.

$\cos(-330°) = \cos 30° = \dfrac{\sqrt{3}}{2}$

c $495° - 360° = 135°$: quadrant 2, related angle 45°, tan negative.

$\tan 495° = -\tan 45° = -1$

C6 **Tutor demo.**

a $145° = 180° - 35°$, quadrant 2 (only sin positive): $\sin 145° = \sin 35° \fallingdotseq 0.57$, $\cos 145° = -\cos 35° \fallingdotseq -0.82$, $\tan 145° = -\tan 35° \fallingdotseq -0.70$

b $325° = 360° - 35°$, quadrant 4 (only cos positive): $\sin 325° = -\sin 35° \fallingdotseq -0.57$, $\cos 325° = \cos 35° \fallingdotseq 0.82$, $\tan 325° = -\tan 35° \fallingdotseq -0.70$

Your turn.

a $125° = 180° - 55°$, quadrant 2: $\sin 125° \fallingdotseq 0.82$, $\cos 125° \fallingdotseq -0.57$, $\tan 125° \fallingdotseq -1.43$

b $235° = 180° + 55°$, quadrant 3 (only tan positive): $\sin 235° \fallingdotseq -0.82$, $\cos 235° \fallingdotseq -0.57$, $\tan 235° \fallingdotseq 1.43$

C7 **Tutor demo.**

a The cosine graph crosses the x-axis at $x = 90°, 270°$. **b** It is below the axis for 90° < x < 270°. **c** The graph is symmetric about $x = 180°$, so the other angle is $360° - 50° = 310°$.

Your turn.

a $x = 0°, 180°, 360°$. **b** 180° < x < 360°. **c** The graph is symmetric about $x = 90°$, so the other angle is $180° - 50° = 130°$.

C8 **Tutor demo.**

a gradient of $AB = \dfrac{0-4}{-4+2} = \dfrac{-4}{-2} = 2$, and $\tan\alpha = 2$ gives $\alpha \div 63.43°$, so the angle of inclination of AB is about 63°.

gradient of $AO = \dfrac{0-4}{0+2} = -2$. The gradient is negative, so α is obtuse:

$\tan(180° - \alpha) = 2$, so $180° - \alpha \div 63.43°$ and $\alpha \div 116.57°$, about 117°.

b $\angle AOB = 180° - 116.57° = 63.43° = \angle ABO$ (straight angle). The base angles are equal, so ΔABO is isosceles.

Your turn.

a gradient of $AB = \dfrac{0-3}{-8+4} = \dfrac{3}{4}$, so $\alpha \div 36.87°$, about 37°.

gradient of $AO = \dfrac{0-3}{0+4} = -\dfrac{3}{4}$, so $180° - \alpha \div 36.87°$ and $\alpha \div 143.13°$, about 143°.

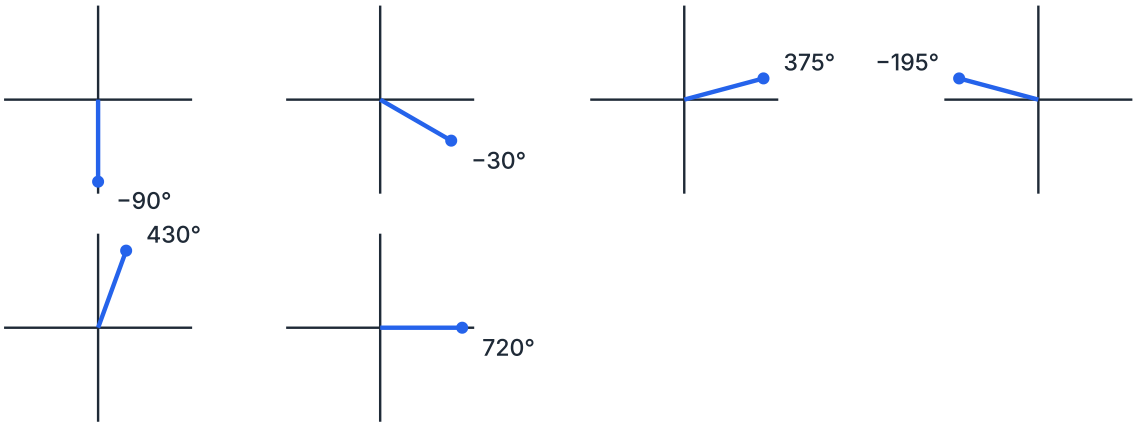
b $\angle AOB = 180° - 143.13° = 36.87° = \angle ABO$, so ΔABO is isosceles.

Homework: Easy

1 Maths Pathway, *Graphs of Trigonometric Functions*, Q1

a) Equivalent to the angle 270° **b)** 330° **c)** 15° **d)** 165° **e)** 70°

f) 0° and 360°



Add or subtract 360°: for example $-195° + 360° = 165°$ and $430° - 360° = 70°$.

2 Maths Pathway, *Graphs of Trigonometric Functions*, Q3

a) 4th quadrant **b)** 1st quadrant **c)** 2nd quadrant **d)** 4th quadrant

e) 3rd quadrant **f)** 4th quadrant **g)** 3rd quadrant **h)** 1st quadrant

Equivalent angles: −45° → 315°, 400° → 40°, −185° → 175°, 640° → 280°, 560° → 200°, −400° → 320°, −98° → 262°, 725° → 5°.

3 Maths Pathway, *The Unit Circle*, Q9

a) Positive **b)** Positive **c)** Positive **d)** Negative **e)** Negative **f)** Positive

g) Negative **h)** Negative

4 Year 11 Advanced Exercise 5D Q5

From the diagram the rays are at a 70°, b 100°, c 140°, d 200°, e 240°, f 340°. Add 360° once, and subtract 360° once and twice.

a 70°, 430°, −290°, −650° **b** 100°, 460°, −260°, −620°

c 140°, 500°, −220°, −580° **d** 200°, 560°, −160°, −520°

e 240°, 600°, −120°, −480° **f** 340°, 700°, −20°, −380°

5 Year 11 Advanced Exercise 5E Q1

a + **b** + **c** − **d** − **e** + **f** − **g** − **h** + **i** − **j** + **k** − **l** −

m − **n** + **o** + **p** −

6 Year 11 Advanced Exercise 5E Q2

a 10° **b** 30° **c** 50° **d** 20° **e** 80° **f** 70° **g** 70° **h** 80° **i** 10°

j 20°

7 Year 11 Advanced Exercise 5E Q3

a $-\tan 50°$ **b** $\cos 50°$ **c** $-\sin 40°$ **d** $\tan 80°$ **e** $-\cos 10°$

f $-\sin 40°$ **g** $-\cos 5°$ **h** $\sin 55°$ **i** $-\tan 35°$ **j** $\sin 85°$ **k** $-\cos 85°$

l $\tan 25°$

8 Year 11 Advanced Exercise 5E Q4

a 0 **b** −1 **c** 0 **d** 0 **e** 1 **f** 1 **g** −1 **h** undefined **i** 0 **j** 0

k undefined **l** 0

9 Year 11 Advanced Exercise 5E Q5

a $\frac{\sqrt{3}}{2}$ **b** $\frac{\sqrt{3}}{2}$ **d** $-\frac{\sqrt{3}}{2}$ **e** $\frac{1}{\sqrt{2}}$ **f** $-\frac{1}{\sqrt{2}}$ **g** $-\frac{1}{\sqrt{2}}$ **h** $\frac{1}{\sqrt{2}}$

i $\frac{1}{\sqrt{3}}$ **j** $-\frac{1}{\sqrt{3}}$ **k** $\frac{1}{\sqrt{3}}$ **l** $-\frac{1}{\sqrt{3}}$

10 Year 11 Advanced Exercise 6B Q6

a $\tan 15° \fallingdotseq 0.27$ **b** $\tan 135° = -1.00$ **c** $\tan 22\frac{1}{2}° \fallingdotseq 0.41$

d $\tan 72° \fallingdotseq 3.08$

11 Year 11 Advanced Exercise 6B Q7

a $\tan\alpha = 1$, $\alpha = 45°$

b $\tan\alpha = -\sqrt{3}$: obtuse, $\tan(180° - \alpha) = \sqrt{3}$, $180° - \alpha = 60°$, $\alpha = 120°$

d $\tan\alpha = \dfrac{1}{\sqrt{3}}$, $\alpha = 30°$

12 Year 11 Advanced Exercise 6B Q8

a $m = \dfrac{5-3}{6-1} = \dfrac{2}{5} < 1$: less **b** $m = \dfrac{4-1}{7-4} = 1$: equal

c $m = \dfrac{5-2}{1+3} = \dfrac{3}{4} < 1$: less **d** $m = \dfrac{-2-4}{-7+3} = \dfrac{3}{2} > 1$: more

Homework: Medium

13 Year 11 Advanced Exercise 5D Q6

a $x = 3,\ y = 4,\ r = 5$: $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$, $\tan\theta = \frac{4}{3}$

b $x = -4,\ y = 3,\ r = 5$: $\sin\theta = \frac{3}{5}$, $\cos\theta = -\frac{4}{5}$, $\tan\theta = -\frac{3}{4}$

c $x = -1,\ y = -2,\ r = \sqrt{5}$: $\sin\theta = -\dfrac{2}{\sqrt{5}}$, $\cos\theta = -\dfrac{1}{\sqrt{5}}$, $\tan\theta = \dfrac{-2}{-1} = 2$

d $x = 12,\ y = -5,\ r = 13$: $\sin\theta = -\frac{5}{13}$, $\cos\theta = \frac{12}{13}$, $\tan\theta = -\frac{5}{12}$

14 Year 11 Advanced Exercise 5E Q6

a $-\cos 60° = -\frac{1}{2}$ **b** $\tan 45° = 1$ **c** $-\sin 30° = -\frac{1}{2}$ **d** $\sin 45° = \dfrac{1}{\sqrt{2}}$

e $\tan 60° = \sqrt{3}$ **f** $-\cos 30° = -\dfrac{\sqrt{3}}{2}$ **g** $-\tan 45° = -1$

h $\cos 60° = \frac{1}{2}$ **i** $-\sin 45° = -\dfrac{1}{\sqrt{2}}$ **j** $-\cos 30° = -\dfrac{\sqrt{3}}{2}$

k $-\sin 30° = -\frac{1}{2}$

15 Year 11 Advanced Exercise 5E Q9

a −60° → 300°: 60° **b** −200° → 160°: 20° **c** −150° → 210°: 30°

d −300° → 60°: 60° **e** 430° → 70°: 70° **f** 530° → 170°: 10°

g 590° → 230°: 50° **h** 680° → 320°: 40°

16 Year 11 Advanced Exercise 5E Q10

a $\cos 60° = \frac{1}{2}$ **b** −120° → 240°: $-\sin 60° = -\frac{\sqrt{3}}{2}$

c −120° → 240°: $\tan 60° = \sqrt{3}$ **d** −315° → 45°: $\sin 45° = \frac{1}{\sqrt{2}}$

e −210° → 150°: $-\tan 30° = -\frac{1}{\sqrt{3}}$ **f** −225° → 135°: $-\cos 45° = -\frac{1}{\sqrt{2}}$

g 420° → 60°: $\tan 60° = \sqrt{3}$ **h** 510° → 150°: $-\cos 30° = -\frac{\sqrt{3}}{2}$

i 495° → 135°: $\sin 45° = \frac{1}{\sqrt{2}}$ **j** 690° → 330°: $-\sin 30° = -\frac{1}{2}$

k 600° → 240°: $-\cos 60° = -\frac{1}{2}$ **l** 585° → 225°: $\tan 45° = 1$

17 Maths Pathway, *Using Symmetry on the Unit Circle*, Q9

a) $\sin(165°) = \sin(15°) = 0.26$ **b)** $\cos(165°) = -\cos(15°) = -0.97$

c) $\tan(315°) = -\tan(45°) = -1$ **d)** $\sin(210°) = -\sin(30°) = -0.5$

e) $\cos(195°) = -\cos(15°) = -0.97$ **f)** $\tan(240°) = \tan(60°) = 1.73$

g) $\sin(300°) = -\sin(60°) = -0.87$ **h)** $\cos(330°) = \cos(30°) = 0.87$

18 Maths Pathway, *Graphs of Trigonometric Functions*, Q5

a) $30° + 360°$ and $150° + 360°$: 390° and 510°

b) 330° → −30°, and −150° has the same sine: −30° and −150°

c) $95° + 360°$ and $265° + 360°$: 455° and 625°

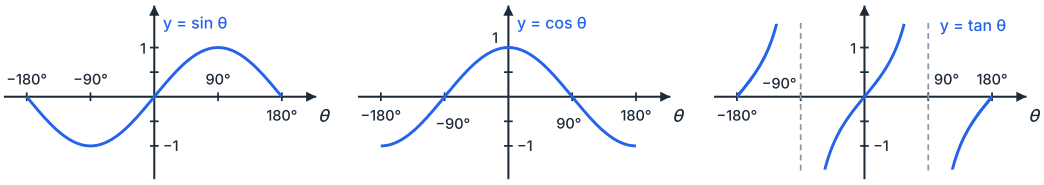
d) $170° + 360°$ and $190° + 360°$: 530° and 550°

19 Maths Pathway, *Graphs of Trigonometric Functions*, Q9

a)

θ	−180°	−90°	0°	90°	180°
$\sin\theta$	0	−1	0	1	0
$\cos\theta$	−1	0	1	0	−1
$\tan\theta$	0	undefined	0	undefined	0

b)



20 Year 11 Advanced Exercise 6B Q17

a $y = 0$: $x = -2$; $x = 0$: $y = 6$. A(−2, 0), B(0, 6), $m = \dfrac{6-0}{0+2} = 3$, $\tan\alpha = 3$, $\alpha \div 72°$

b $y = 0$: $x = 2$; $x = 0$: $y = 1$. A(2, 0), B(0, 1), $m = \dfrac{1-0}{0-2} = -\frac{1}{2}$, $\tan(180° - \alpha) = \frac{1}{2}$, $\alpha \div 180° - 26.57° \div 153°$

c $y = 0$: $x = -4$; $x = 0$: $y = -3$. A(−4, 0), B(0, −3), $m = \dfrac{-3-0}{0+4} = -\frac{3}{4}$, $\alpha \div 180° - 36.87° \div 143°$

d $y = 0$: $x = 3$; $x = 0$: $y = -2$. A(3, 0), B(0, −2), $m = \dfrac{-2-0}{0-3} = \frac{2}{3}$, $\tan\alpha = \frac{2}{3}$, $\alpha \div 34°$

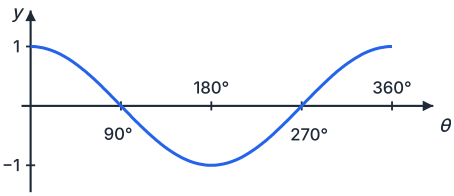
21 Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q13

Drop a perpendicular from P to the x-axis. The right-angled triangle has opposite side 5 and adjacent side 4 from θ (the gradient of OP is $\frac{5}{4}$).

$\tan\theta = \dfrac{5}{4}$, so $\theta = \tan^{-1}\frac{5}{4} \div 51.34° \div 51°$. The answer is C. Option A (38°) and B (39°) come from $\tan^{-1}\frac{4}{5} \div 38.66°$, which is the angle with the y-axis; D is 51.34° rounded up.

22 Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q6(c)

(i) Mark the key points first: $(0°, 1)$, $(90°, 0)$, $(180°, -1)$, $(270°, 0)$, $(360°, 1)$. Join them with a smooth wave, labelling the axes, the maximum 1 and the minimum −1.



(ii) The graph crosses the θ-axis where $\cos\theta = 0$: $\theta = 90°$ or 270°.

Homework: Hard

23 Year 11 Advanced Exercise 5D Q8

Readings from a graph are approximate; the book's answers are given.

a i 0.5 **ii** −0.5 **iii** 0.95 **iv** 0.95 **v** 0.59 **vi** 0.81 **vii** −0.89 **viii** 0.45 **ix** −0.81 **x** 0.59

b i 30°, 150° **ii** 120°, 240° **iii** 64°, 116° **iv** 53°, 307° **v** 53°, 127° **vi** 143°, 217° **vii** 204°, 336° **viii** 107°, 253°

c The two curves cross at 45°, 225°.

24 Year 11 Advanced Exercise 5E Q12

a $\sin 155^\circ = \sin 25^\circ \fallingdotseq 0.42$ **b** $\cos 205^\circ = -\cos 25^\circ \fallingdotseq -0.91$

c $\cos 335^\circ = \cos 25^\circ \fallingdotseq 0.91$ **d** $\sin 335^\circ = -\sin 25^\circ \fallingdotseq -0.42$

e $\sin 205^\circ - \cos 155^\circ = -\sin 25^\circ - (-\cos 25^\circ) \fallingdotseq -0.42 + 0.91 = 0.49$

f $385^\circ \to 25^\circ$ and $515^\circ \to 155^\circ$:

$\cos 385^\circ - \sin 515^\circ = \cos 25^\circ - \sin 25^\circ \fallingdotseq 0.91 - 0.42 = 0.49$

25 Year 11 Advanced Exercise 5E Q14

a $\theta = 150^\circ$, $2\theta = 300^\circ$. LHS $= \sin 300^\circ = -\sin 60^\circ = -\dfrac{\sqrt{3}}{2}$. RHS

$= 2\sin 150^\circ \cos 150^\circ = 2 \times \frac{1}{2} \times \left(-\dfrac{\sqrt{3}}{2}\right) = -\dfrac{\sqrt{3}}{2}$. So LHS = RHS.

b $\theta = 225^\circ$, $3\theta = 675^\circ \to 315^\circ$. LHS $= \cos 315^\circ = \cos 45^\circ = \dfrac{1}{\sqrt{2}}$. With

$\cos 225^\circ = -\dfrac{1}{\sqrt{2}}$ and $\left(-\dfrac{1}{\sqrt{2}}\right)^3 = -\dfrac{1}{2\sqrt{2}}$,

RHS $= 4 \times \left(-\dfrac{1}{2\sqrt{2}}\right) - 3 \times \left(-\dfrac{1}{\sqrt{2}}\right) = -\dfrac{2}{\sqrt{2}} + \dfrac{3}{\sqrt{2}} = \dfrac{1}{\sqrt{2}}$. So LHS = RHS.

c $A + B = 540^\circ \to 180^\circ$. LHS $= \sin 540^\circ = \sin 180^\circ = 0$. With

$\sin 300^\circ = -\dfrac{\sqrt{3}}{2}$, $\cos 300^\circ = \frac{1}{2}$, $\sin 240^\circ = -\dfrac{\sqrt{3}}{2}$, $\cos 240^\circ = -\frac{1}{2}$,

RHS $= \left(-\dfrac{\sqrt{3}}{2}\right)\left(-\dfrac{1}{2}\right) + \left(\dfrac{1}{2}\right)\left(-\dfrac{\sqrt{3}}{2}\right) = \dfrac{\sqrt{3}}{4} - \dfrac{\sqrt{3}}{4} = 0$. So LHS = RHS.

26 Year 11 Advanced Exercise 5E Q15

a $-\sin\theta$ **b** $\cos\theta$ **c** $-\tan\theta$ **e** $\sin\theta$ **f** $-\sin\theta$ **g** $-\cos\theta$ **h** $\tan\theta$

Place θ as if acute: $-\theta$ and $360^\circ - \theta$ are in quadrant 4, $180^\circ - \theta$ in quadrant 2, $180^\circ + \theta$ in quadrant 3.

27 Year 11 Advanced Exercise 6B Q20

The line slopes upwards, so its angle of inclination is $90^\circ -$ (its acute angle with the y-axis).

a $\tan 75^\circ \fallingdotseq 3.73$ **b** $\tan 45^\circ = 1$ **c** $\tan 67\frac{1}{2}^\circ \fallingdotseq 2.41$ **d** $\tan 18^\circ \fallingdotseq 0.32$

28 Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q8

The graph passes through $(0, 0)$, $(180, 0)$ and $(360, 0)$, so it is a sine curve, not a cosine curve (C and D are 1 or −1 at $x = 0$). At $x = 90$ the graph is at −1, but $\sin 90^\circ = 1$, so it is $y = -\sin x^\circ$. Since $\sin(-x)^\circ = -\sin x^\circ$ (Hard Question 26(a)), the answer is B.

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T1 | Angles of any size

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q4**

$\sin\theta > 0$ in quadrants 1 and 2 (A and S). $\cos\theta < 0$ in quadrants 2 and 3. Both hold only in quadrant 2. The answer is B.

PP 2 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Part A, Q12**

An obtuse angle is in quadrant 2, where $\tan\theta = -\tan(180° - \theta)$. So $\tan\theta = -\tan 50°$ when $180° - \theta = 50°$, that is, $\theta = 130°$. The answer is A. The others give $-\tan 40°$, $-\tan 35°$ and $-\tan 25°$.

PP 3 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Part A, Q13**

(A) cos positive (quadrants 1, 4) and sin negative (quadrants 3, 4): quadrant 4, correct. (B) cos positive (1, 4) and tan positive (1, 3): quadrant 1, *not* quadrant 3. (C) tan negative (2, 4) and cos negative (2, 3): quadrant 2, correct. (D) sin positive (1, 2) and tan positive (1, 3): quadrant 1, correct. The incorrect statement is B.

PP 4 **Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Multiple Choice, Q3**

Line k passes through $(2, 0)$ and $(0, -4)$, so its gradient is $\dfrac{0-(-4)}{2-0} = \dfrac{4}{2} = 2$. The angle Ø is its angle of inclination, so $\tan Ø = 2$ and $Ø = \tan^{-1} 2 \doteqdot 63.43° \doteqdot 63°$. The answer is C. Options A and B (26° and 27°) come from $\tan^{-1}\frac{1}{2} \doteqdot 26.57°$, which is the angle between the line and the y-axis; D is 63.43° rounded up.

PP 5 **The Scots College, 2024, Year 10 Assessment Task 4, Mathematics 5.3, Q28(a)**

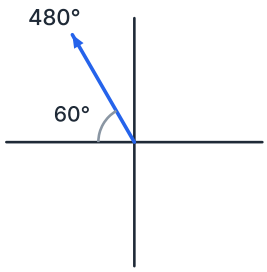
286° is between 270° and 360°, so it is in quadrant 4, where cos is positive. Its related acute angle is $360° - 286° = 74°$, so $\cos 286° = +\cos 74°$. Since θ is acute, $\theta = 74°$.

PP 6 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q14**

The line $y = 4x - 5$ has gradient 4, so its angle of inclination α satisfies $\tan\alpha = 4$. The gradient is positive, so α is acute: $\alpha = \tan^{-1} 4 \doteqdot 75.96°$. Then $0.96° \times 60 \doteqdot 57.8'$, which rounds to $58'$, so $\alpha \doteqdot 75°58'$.

PP 7 **Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q1(h)**

$480° = 360° + 120°$, so after one full revolution the ray turns a further 120°, into the second quadrant. The quadrant diagram shows the ray at 120° from the positive x-axis. The related angle is the acute angle with the x-axis: $180° - 120° = 60°$.

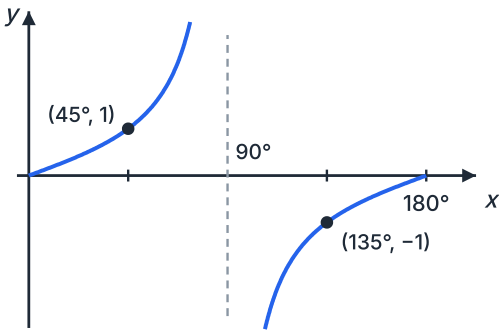


PP 8 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part C, (f)**

Here $x = -3$, $y = 4$ and $r = OP = 5$ (check: $\sqrt{9+16} = 5$). So $\sin\theta = \dfrac{y}{r} = \dfrac{4}{5}$ and $\tan\theta = \dfrac{y}{x} = \dfrac{4}{-3} = -\dfrac{4}{3}$. There is no need to find θ itself.

PP 9 **North Sydney Girls High School, 2022, Year 10 Mathematics Yearly Examination, Q16(c)**

$\tan 0° = 0$, $\tan 45° = 1$, $\tan 135° = -1$, $\tan 180° = 0$, and $\tan 90°$ is undefined. As x increases towards 90°, $\tan x$ increases without bound; just after 90° it is very large and negative. So draw a vertical asymptote at $x = 90°$, a rising curve from $(0°, 0)$ through $(45°, 1)$ up along the asymptote, and a rising curve from below the axis near the asymptote through $(135°, -1)$ to $(180°, 0)$.



PP 10 **North Sydney Boys High School, 2020, Stage 5 Mathematics Year 10 Yearly Examination, Q2(b)**

$\sin 150° = \sin 30° = \frac{1}{2}$ (quadrant 2). $\cos 300° = \cos 60° = \frac{1}{2}$ (quadrant 4).

$750° - 720° = 30°$, so $\tan 750° = \tan 30° = \dfrac{1}{\sqrt{3}}$. $\tan 120° = -\tan 60° = -\sqrt{3}$ (quadrant 2).

$$\frac{\sin 150° + \cos 300°}{\tan 750° - \tan 120°} = \frac{\frac{1}{2} + \frac{1}{2}}{\frac{1}{\sqrt{3}} - (-\sqrt{3})} = \frac{1}{\frac{1}{\sqrt{3}} + \sqrt{3}}$$

$$= \frac{1}{\frac{1+3}{\sqrt{3}}} = \frac{\sqrt{3}}{4}$$

The exact value is $\dfrac{\sqrt{3}}{4}$.

PP 11 **Sydney Technical High School, 2020, Year 10 Mathematics Assessment 4, Q7(c)**

(i) $\cos 45° = \dfrac{1}{\sqrt{2}}$ (exact value).

(ii) 240° is in quadrant 3, related angle 60°, sin negative: $\sin 240° = -\sin 60° = -\dfrac{\sqrt{3}}{2}$.

(iii) $420° - 360° = 60°$, so $\tan 420° = \tan 60° = \sqrt{3}$, and $\tan^2(420°) = \left(\sqrt{3}\right)^2 = 3$.

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T2 | Exact values and trigonometric equations

Lesson Examples

C1 **Tutor demo.**

a $\cos^2 45° + \sin 30° = \left(\frac{1}{\sqrt{2}}\right)^2 + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$

b $\tan 30° \times \tan 60° = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$

c x is opposite 45° and 8 is opposite 120°. $\frac{x}{\sin 45°} = \frac{8}{\sin 120°}$, so $x = \frac{8 \sin 45°}{\sin 120°}$.

Here $\sin 45° = \frac{1}{\sqrt{2}}$ and $\sin 120° = +\sin 60° = \frac{\sqrt{3}}{2}$ (quadrant 2, related angle 60°).

$$x = 8 \times \frac{1}{\sqrt{2}} \times \frac{2}{\sqrt{3}} = \frac{16}{\sqrt{6}} = \frac{16\sqrt{6}}{6} = \frac{8\sqrt{6}}{3}$$

Your turn.

a $\sin^2 60° + \cos 60° = \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$

b $\cos 30° \times \tan 30° = \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2}$

c $\frac{x}{\sin 30°} = \frac{6}{\sin 120°}$, so $x = \frac{6 \sin 30°}{\sin 120°} = 6 \times \frac{1}{2} \times \frac{2}{\sqrt{3}} = \frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$

C2 **Tutor demo.**

a $\sin 40° = \cos(90° - 40°) = \cos 50°$, so $x = 50°$.

b $\frac{\cos(90° - \theta)}{\cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

c $\cos 65° = \sin(90° - 65°) = \sin 25°$, so $\sin 25° - \cos 65° = 0$.

Your turn.

a $\cos 15° = \sin(90° - 15°) = \sin 75°$, so $x = 75°$.

b $\sin(90° - \theta) \tan \theta = \cos \theta \times \frac{\sin \theta}{\cos \theta} = \sin \theta$

c $\sin 58° = \cos(90° - 58°) = \cos 32°$, so $\cos 32° - \sin 58° = 0$.

C3 **Tutor demo.**

a $4\sin^2\theta + 4\cos^2\theta = 4(\sin^2\theta + \cos^2\theta) = 4 \times 1 = 4$ (Box 21).

b From Box 21, $1 - \sin^2\theta = \cos^2\theta$.

$$\begin{aligned}\text{LHS} &= (1 - \sin^2\theta)\tan^2\theta \\ &= \cos^2\theta \times \frac{\sin^2\theta}{\cos^2\theta} \qquad \text{(Boxes 21 and 20)} \\ &= \sin^2\theta \\ &= \text{RHS}.\end{aligned}$$

Your turn.

a $3 - 3\cos^2\theta = 3(1 - \cos^2\theta) = 3\sin^2\theta$ (Box 21: $1 - \cos^2\theta = \sin^2\theta$).

b

$$\begin{aligned}\text{LHS} &= \tan\theta\,(1 - \sin^2\theta) \\ &= \frac{\sin\theta}{\cos\theta} \times \cos^2\theta \qquad \text{(Boxes 20 and 21)} \\ &= \sin\theta\cos\theta \\ &= \text{RHS}.\end{aligned}$$

C4 **Tutor demo.**

a $\cos\theta$ is positive, so θ is in quadrant 1 or 4. Take $x = 2$, $r = 7$: $y^2 = 7^2 - 2^2 = 45$, so $y = \pm\sqrt{45} = \pm 3\sqrt{5}$, and $\sin\theta = \pm\dfrac{3\sqrt{5}}{7}$.

b $\tan\theta$ negative: quadrant 2 or 4. With quadrant 1 or 4, only quadrant 4, where $y < 0$: $\sin\theta = -\dfrac{3\sqrt{5}}{7}$.

Your turn.

a $\cos\theta$ is negative, so quadrant 2 or 3. Take $x = -1$, $r = 3$: $y^2 = 9 - 1 = 8$, $y = \pm 2\sqrt{2}$, and $\sin\theta = \pm\dfrac{2\sqrt{2}}{3}$.

b $\tan\theta$ positive: quadrant 1 or 3. With quadrant 2 or 3, only quadrant 3, where $y < 0$: $\sin\theta = -\dfrac{2\sqrt{2}}{3}$.

C5 **Tutor demo.** From the graph of $y = \cos x$ over two periods:

a it touches $y = 1$ at $x = 0°,\ 360°$ or $720°$. **b** it crosses the x-axis at $x = 90°,\ 270°,\ 450°$ or $630°$.

Your turn.

a $x = 90°$ or $450°$ **b** $x = 0°,\ 180°,\ 360°,\ 540°$ or $720°$

C6 **Tutor demo.**

a $\cos x$ negative: quadrant 2 or 3. Related angle: $\cos\alpha = \dfrac{\sqrt{3}}{2}$, $\alpha = 30°$. In $-180° \le x \le 180°$: quadrant 2 gives $180° - 30° = 150°$, quadrant 3 gives $-150°$. So $x = -150°$ or $150°$.

b $\sin x$ positive: quadrant 1 or 2. Related angle $\sin^{-1} 0.6 \fallingdotseq 36.87°$. $x \fallingdotseq 36.87°$ or $180° - 36.87° = 143.13°$, so $x \fallingdotseq 37°$ or $143°$.

Your turn.

a $\tan x$ negative: quadrant 2 or 4, related angle 45°. In $-180° \le x \le 180°$: $x = -45°$ or $135°$.

b $\cos x$ negative: quadrant 2 or 3. Related angle $\cos^{-1} 0.35 \fallingdotseq 69.51°$ (enter 0.35, not −0.35). $x \fallingdotseq 180° - 69.51° = 110.49°$ or $180° + 69.51° = 249.51°$, so $x \fallingdotseq 110°$ or $250°$.

C7 **Tutor demo.** Dividing by $\cos x$ (it is not 0, or $\sin x$ would be 0 too), $\tan x = 2$. Positive: quadrant 1 or 3. Related angle $\tan^{-1} 2 \fallingdotseq 63.43°$, so $x \fallingdotseq 63.43°$ or $180° + 63.43° = 243.43°$, and $x \fallingdotseq 63°$ or $243°$.

Your turn. Dividing by $\sqrt{3}\cos x$, $\tan x = \dfrac{1}{\sqrt{3}}$. Quadrant 1 or 3, related angle 30°, so $x = 30°$ or $210°$.

C8 **Tutor demo.**

a Let $u = 2x$; $0° \le u \le 720°$; $\cos u = \frac{1}{2}$: quadrant 1 or 4, related angle 60°. $u = 60°,\ 300°,\ 420°,\ 660°$, so $x = 30°,\ 150°,\ 210°$ or $330°$.

b Let $u = x - 60°$; $-60° \le u \le 300°$; $\tan u = 1$: $u = 45°$ or $225°$ ($-135°$ is outside the domain). So $x = 105°$ or $285°$.

Your turn.

a $u = 2x$, $0° \le u \le 720°$, $\sin u = \dfrac{\sqrt{3}}{2}$: $u = 60°,\ 120°,\ 420°,\ 480°$, so $x = 30°,\ 60°,\ 210°$ or $240°$.

b $u = x + 45°$, $45° \le u \le 405°$, $\cos u = -\frac{1}{2}$: quadrant 2 or 3, related angle 60°: $u = 120°$ or $240°$ ($480°$ is too big). So $x = 75°$ or $195°$.

Homework: Easy

1 Year 11 Advanced Exercise 5A Q10

a $\dfrac{\sqrt{3}}{2}$ **b** $\dfrac{1}{\sqrt{3}}$ **c** $\dfrac{1}{\sqrt{2}}$

2 Year 11 Advanced Exercise 5A Q17

a $\dfrac{1}{\sqrt{2}} \times \dfrac{1}{\sqrt{2}} + \dfrac{1}{2} = \dfrac{1}{2} + \dfrac{1}{2} = 1$ **b** $\dfrac{\sqrt{3}}{2} \times \dfrac{\sqrt{3}}{2} - \dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{3}{4} - \dfrac{1}{4} = \dfrac{1}{2}$

c $1 + \left(\sqrt{3}\right)^2 = 1 + 3 = 4$

3 Year 11 Advanced Exercise 5G Q1

a $\sin 16^\circ \fallingdotseq 0.2756$ and $\cos 74^\circ \fallingdotseq 0.2756$: both are 0.2756..., so they are equal (16° and 74° are complementary).

d $\sin 23^\circ \fallingdotseq 0.3907$ and $\cos 23^\circ \fallingdotseq 0.9205$, so $\sin^2 23^\circ + \cos^2 23^\circ \fallingdotseq 0.1527 + 0.8473 = 1.0000$. (Use the stored values, not the rounded ones, and the calculator shows exactly 1.)

4 Year 11 Advanced Exercise 5G Q4

a

$$\begin{aligned} \text{LHS} &= \tan\theta\cos\theta \\ &= \frac{\sin\theta}{\cos\theta} \times \cos\theta \qquad \text{(Box 20)} \\ &= \sin\theta \\ &= \text{RHS.} \end{aligned}$$

5 Year 11 Advanced Exercise 5G Q5

a $\cos\theta$ **d** $\dfrac{\cos(90^\circ - \phi)}{\sin(90^\circ - \phi)} = \dfrac{\sin\phi}{\cos\phi} = \tan\phi$

6 Year 11 Advanced Exercise 5G Q6 a and Q7 a

6a $\sin^2\alpha + \cos^2\alpha = 1$ (Box 21). **7a** From Box 21, $\cos^2\beta = 1 - \sin^2\beta$, so $1 - \sin^2\beta = \cos^2\beta$.

7 Year 11 Advanced Exercise 5F Q1

a $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$, $\tan\theta = \frac{15}{8}$

b $\sin\theta = \frac{4}{5}$, $\cos\theta = -\frac{3}{5}$, $\tan\theta = -\frac{4}{3}$

c $\sin\theta = -\frac{7}{25}$, $\cos\theta = -\frac{24}{25}$, $\tan\theta = \frac{7}{24}$

d $\sin\theta = -\frac{21}{29}$, $\cos\theta = \frac{20}{29}$, $\tan\theta = -\frac{21}{20}$

For example in **b**, $x = -6, y = 8, r = 10$: $\sin\theta = \frac{8}{10}$, $\cos\theta = \frac{-6}{10}$, $\tan\theta = \frac{8}{-6}$.

8 Year 11 Advanced Exercise 5F Q5

a Q1: $x = 1$, $r = 3$, $y = \sqrt{8} = 2\sqrt{2}$; $\tan\theta = 2\sqrt{2}$ **b** Q2: $x = -4$, $r = 5$, $y = 3$; $\tan\theta = -\frac{3}{4}$

c Reflex with cos positive: Q4. $x = 1$, $r = 2$, $y = -\sqrt{3}$; $\sin\theta = -\dfrac{\sqrt{3}}{2}$ **d** Reflex with tan negative: Q4. $x = 3$, $y = -2$, $r = \sqrt{13}$; $\cos\theta = \dfrac{3}{\sqrt{13}}$

e Q2: $x = -40$, $r = 41$, $y = 9$; $\sin\theta = \frac{9}{41}$ **f** Q1: $y = 1$, $r = \sqrt{5}$, $x = 2$; $\tan\theta = \frac{1}{2}$

9 Year 11 Advanced Exercise 5H Q1

a $60°$ or $120°$ **c** $45°$ or $225°$ **d** $60°$ or $240°$ **e** $135°$ or $225°$

f $120°$ or $300°$ **h** $150°$ or $210°$

10 Year 11 Advanced Exercise 5H Q2

a $90°$ **b** $0°$ or $360°$ **c** $90°$ or $270°$ **d** $180°$ **e** $0°,\ 180°$ or $360°$

f $270°$

11 Year 11 Advanced Exercise 5H Q3

a related $64.62°$; Q1, Q4: $65°$ or $295°$ **b** related $7.09°$; Q1, Q2: $7°$ or $173°$

c related $81.87°$; Q1, Q3: $82°$ or $262°$ **d** related $41.81°$; Q3, Q4: $222°$ or $318°$

e related $65.77°$; Q2, Q4: $114°$ or $294°$ **f** related $39.65°$; Q2, Q3: $140°$ or $220°$

12 Maths Pathway, *Solving Trigonometric Equations*, Q8

a) $\sin(x) = \frac{3}{4}$: $x = 48.59°$ or $131.41°$

b) $\tan(x) = 55$: $x = 88.96°$ or $268.96°$ **c)** $\cos(x) = -\frac{1}{2}$: $x = 120°$ or $240°$

d) $\tan(x) = -\frac{12}{5}$: $x = 112.62°$ or $292.62°$

Homework: Medium

13 Year 11 Advanced Exercise 5G Q8

b $\dfrac{\sin^2\beta}{\cos^2\beta} = \left(\dfrac{\sin\beta}{\cos\beta}\right)^2 = \tan^2\beta$ (Box 20).

14 Year 11 Advanced Exercise 5G Q11

a

$$\begin{aligned}\text{LHS} &= 1 - \sin^2\theta \qquad \text{(difference of squares)}\\ &= \cos^2\theta \qquad \text{(Box 21)}\\ &= \text{RHS.}\end{aligned}$$

b

$$\begin{aligned}\text{LHS} &= \cos^2\alpha + \tan^2\alpha\cos^2\alpha \qquad \text{(expand)}\\ &= \cos^2\alpha + \frac{\sin^2\alpha}{\cos^2\alpha} \times \cos^2\alpha \qquad \text{(Box 20)}\\ &= \cos^2\alpha + \sin^2\alpha\\ &= 1 \qquad \text{(Box 21)}\\ &= \text{RHS.}\end{aligned}$$

c

$$\begin{aligned}\text{LHS} &= \sin^2 A + 2\sin A\cos A + \cos^2 A \qquad \text{(expand the square)}\\ &= (\sin^2 A + \cos^2 A) + 2\sin A\cos A\\ &= 1 + 2\sin A\cos A \qquad \text{(Box 21)}\\ &= \text{RHS.}\end{aligned}$$

d

$$\begin{aligned}\text{LHS} &= \cos^2 x - \sin^2 x\\ &= (1 - \sin^2 x) - \sin^2 x \qquad \text{(Box 21)}\\ &= 1 - 2\sin^2 x\\ &= \text{RHS.}\end{aligned}$$

f

$$\begin{aligned}\text{LHS} &= 3\cos^2\theta - 2\\ &= 3(1 - \sin^2\theta) - 2 \qquad \text{(Box 21)}\\ &= 3 - 3\sin^2\theta - 2\\ &= 1 - 3\sin^2\theta\\ &= \text{RHS.}\end{aligned}$$

i

$$\begin{aligned}\text{LHS} &= \cos^4 x + \cos^2 x\sin^2 x\\ &= \cos^2 x(\cos^2 x + \sin^2 x) \qquad \text{(common factor)}\\ &= \cos^2 x \times 1 \qquad \text{(Box 21)}\\ &= \cos^2 x\\ &= \text{RHS.}\end{aligned}$$

15 Year 11 Advanced Exercise 5F Q6

a $r = \sqrt{10}$: $\pm\dfrac{1}{\sqrt{10}}$ **c** $x = \pm 4$: $\pm\frac{4}{5}$ **d** $y = \pm\sqrt{5}$: $\pm\dfrac{\sqrt{5}}{2}$

e $x = \pm 5$: $\pm\frac{12}{5}$ **f** $r = \sqrt{7}$: $\pm\dfrac{\sqrt{3}}{\sqrt{7}}$

16 Year 11 Advanced Exercise 5F Q7

a Q4: $-\frac{3}{4}$ **b** Q2, $r = 17$: $-\frac{15}{17}$ **c** Q2, $x = -\sqrt{15}$: $-\dfrac{\sqrt{15}}{4}$

d Q1, $r = 37$: $\frac{35}{37}$ **e** Q4, $x = 20$: $-\frac{21}{20}$ **f** Q2, $y = \sqrt{11}$: $\dfrac{\sqrt{11}}{6}$

17 Year 11 Advanced Exercise 5H Q4

a $\alpha \doteqdot 5°44'$ or $174°16'$ **b** $\alpha \doteqdot 95°44'$ or $264°16'$

e no solutions, because $\sin\alpha$ can never be greater than 1

18 Year 11 Advanced Exercise 5H Q5

a related $16°42'$: $x \doteqdot -16°42'$ or $163°18'$ **b** $x = -90°$ or $90°$

d related $44°26'$: $x \doteqdot -135°34'$ or $-44°26'$

19 Year 11 Advanced Exercise 5H Q6

a $\cos\theta = \frac{1}{2}$: $\theta = 60°,\ 300°,\ 420°$ or $660°$

d related angle $\tan^{-1}(\sqrt{2} - 1) = 22°30'$; quadrant 1 or 3, twice round:
$\theta = 22°30',\ 202°30',\ 382°30'$ or $562°30'$

20 Maths Pathway, *Solving Trigonometric Equations*, Q9

a) $\sin(x) = \frac{7}{8}$, $\sin^{-1}\frac{7}{8} = 61.04°$; $180° - 61.04° + 360° = 478.96°$

b) $\tan(x) = 12$, $\tan^{-1}12 = 85.24°$; $85.24° + 360° = 445.24°$

c) $\sin(x) = -\frac{4}{7}$, related $34.85°$; $180° + 34.85° = 214.85°$

d) $\cos(x) = \frac{6}{11}$, $\cos^{-1}\frac{6}{11} = 56.94°$; $x = -56.94°$

21 Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Short Answer Q10

$\tan\theta$ is negative, so θ is in quadrant 2 or 4. The related angle is 30°, because $\tan 30° = \dfrac{1}{\sqrt{3}}$. So $\theta = 150°$ or $330°$ ($180° - 30°$ and $360° - 30°$).

22 Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q16

$\cos(90 - \theta) = \sin\theta$ and $\sin(90 - \theta) = \cos\theta$, so the expression is $\dfrac{\sin^2\theta + \cos\theta\sin\theta}{\cos^2\theta + \sin\theta\cos\theta} = \dfrac{\sin\theta(\sin\theta + \cos\theta)}{\cos\theta(\cos\theta + \sin\theta)} = \dfrac{\sin\theta}{\cos\theta} = \tan\theta$. The answer is C.

23 North Sydney Girls High School, 2022, Year 10 Mathematics Yearly Examination, Q19(c)

$\sin x$ is negative, so x is in quadrant 3 or 4. The related angle is 60°, because $\sin 60° = \dfrac{\sqrt{3}}{2}$. So $x = 180° + 60°$ or $360° - 60°$, that is, $x = 240°$ or $300°$ (the school's answer).

24 Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section II Q33

a) By Pythagoras' theorem, $AC^2 = (\sqrt{2})^2 + 2^2 = 2 + 4 = 6$, so $AC = \sqrt{6}$ cm.

b) $\cos A = \dfrac{\text{adj}}{\text{hyp}} = \dfrac{AB}{AC} = \dfrac{\sqrt{2}}{\sqrt{6}} = \dfrac{1}{\sqrt{3}} = \dfrac{1}{\sqrt{3}} \times \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$.

c) $\sin A = \dfrac{\text{opp}}{\text{hyp}} = \dfrac{BC}{AC} = \dfrac{2}{\sqrt{6}}$.

$$\begin{aligned}\text{LHS} &= \left(\frac{2}{\sqrt{6}}\right)^2 + \left(\frac{\sqrt{3}}{3}\right)^2 \\ &= \frac{4}{6} + \frac{3}{9} \\ &= \frac{2}{3} + \frac{1}{3} \\ &= 1 \\ &= \text{RHS}.\end{aligned}$$

25 Fort Street High School, 2022, Year 10 Common Task 2, Mathematics, Section D Question 2

By the complementary identities, $\cos(90° - \theta) = \sin\theta$ and $\sin(90° - \theta) = \cos\theta$.

$$\begin{aligned}\text{LHS} &= \frac{\cos(90° - \theta)\sin\theta}{\sin(90° - \theta)\cos\theta} \\ &= \frac{\sin\theta\sin\theta}{\cos\theta\cos\theta} \\ &= \frac{\sin^2\theta}{\cos^2\theta} \\ &= \left(\frac{\sin\theta}{\cos\theta}\right)^2 \\ &= \tan^2\theta \qquad \text{(Box 20)} \\ &= \text{RHS}.\end{aligned}$$

Homework: Hard

26 Year 11 Advanced Exercise 5A Q15

a A right-angled triangle with the side opposite α equal to $\sqrt{5}$ and the side adjacent to α equal to 2. **b** Hypotenuse $= \sqrt{(\sqrt{5})^2 + 2^2} = \sqrt{9} = 3$. **c** $\sin\alpha = \dfrac{\sqrt{5}}{3}$, $\cos\alpha = \dfrac{2}{3}$.

d

$$\begin{aligned}\text{LHS} &= \sin^2\alpha + \cos^2\alpha \\ &= \left(\frac{\sqrt{5}}{3}\right)^2 + \left(\frac{2}{3}\right)^2 \\ &= \frac{5}{9} + \frac{4}{9} \\ &= 1 \\ &= \text{RHS}.\end{aligned}$$

27 Year 11 Advanced Exercise 5A Q18

b LHS $= 2 \times \frac{1}{2} \times \dfrac{\sqrt{3}}{2} = \dfrac{\sqrt{3}}{2} = \sin 60° =$ RHS.

c LHS $= \left(\frac{1}{2}\right)^2 - \left(\dfrac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} =$ RHS.

d $\tan^2 30° = \frac{1}{3}$, so LHS $= \frac{2}{\sqrt{3}} \div \frac{2}{3} = \frac{2}{\sqrt{3}} \times \frac{3}{2} = \frac{3}{\sqrt{3}} = \sqrt{3} = \tan 60° =$ RHS.

28 North Sydney Boys High School, 2022, Stage 5.3 Mathematics Year 10 Assessment Task 4, Section X Q24(a)–(d)

a) In ΔDBC, CD is opposite θ: $CD^2 = DB^2 + BC^2 - 2 \times DB \times BC \times \cos\theta$, so $(6\sqrt{7})^2 = x^2 + (3x)^2 - 2 \times x \times 3x\cos\theta$, that is, $252 = 10x^2 - 6x^2\cos\theta$. Hence $6x^2\cos\theta = 10x^2 - 252$ and $\cos\theta = \dfrac{10x^2 - 252}{6x^2} = \dfrac{5x^2 - 126}{3x^2}$.

b) $\angle ABC = 90°$, so $\angle ABD = 90° - \theta$. ΔADB is isosceles ($AD = DB = x$), so the perpendicular from D to AB meets AB at its midpoint M, and $BM = 3\sqrt{3}$. In right-angled ΔDMB, $\cos(90° - \theta) = \dfrac{BM}{DB} = \dfrac{3\sqrt{3}}{x}$. Since $\cos(90° - \theta) = \sin\theta$ and $3\sqrt{3} = \sqrt{27}$, $\sin\theta = \dfrac{\sqrt{27}}{x}$. (The school also accepts the cosine rule in ΔADB.)

c) Substitute (a) and (b) into $\sin^2\theta + \cos^2\theta = 1$: $\dfrac{27}{x^2} + \dfrac{(5x^2 - 126)^2}{9x^4} = 1$. Multiply both sides by $9x^4$: $243x^2 + 25x^4 - 1260x^2 + 15876 = 9x^4$, so $16x^4 - 1017x^2 + 15876 = 0$.

d) This is a quadratic in x^2:

$$x^2 = \frac{1017 \pm \sqrt{1017^2 - 4 \times 16 \times 15876}}{2 \times 16} = \frac{1017 \pm \sqrt{18225}}{32} = \frac{1017 \pm 135}{32},$$

so $x^2 = 36$ or $x^2 = \dfrac{441}{16}$. A length is positive, so $x = 6$ or $x = \dfrac{21}{4}$ (the school's answer). Both give a triangle that fits the diagram.

29 Year 11 Advanced Exercise 5G Q13

c

$$\begin{aligned}\text{LHS} &= \sin^4 A - \cos^4 A \\ &= (\sin^2 A - \cos^2 A)(\sin^2 A + \cos^2 A) \qquad \text{(difference of squares)} \\ &= (\sin^2 A - \cos^2 A) \times 1 \qquad \text{(Box 21)} \\ &= \sin^2 A - \cos^2 A \\ &= \text{RHS}.\end{aligned}$$

30 Year 11 Advanced Exercise 5F Q11

θ is obtuse, so it is in quadrant 2. Take $y = p$, $r = q$; then $x^2 = q^2 - p^2$ and $x = -\sqrt{q^2 - p^2}$ (quadrant 2). So $\cos\theta = -\dfrac{\sqrt{q^2 - p^2}}{q}$, $\tan\theta = -\dfrac{p}{\sqrt{q^2 - p^2}}$.

31 Year 11 Advanced Exercise 5H Q9

Divide by $\cos\theta$ each time (it cannot be 0, because then $\sin\theta$ would be 0 as well).

a $\tan\theta = 1$: $45°$ or $225°$ **b** $\tan\theta = -1$: $135°$ or $315°$

c $\tan\theta = \sqrt{3}$: $60°$ or $240°$ **d** $\tan\theta = -\dfrac{1}{\sqrt{3}}$: $150°$ or $330°$

32 Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q12

ΔABX: $AX = BX = 12\sin 45° = 6\sqrt{2}$. $BCYX$ is a rectangle: $XY = 4\sqrt{2}$, $CY = 6\sqrt{2}$. $\angle YCD = 150° - 90° = 60°$, so in ΔCYD, $YD = 6\sqrt{2}\tan 60° = 6\sqrt{6}$. $AD = 6\sqrt{2} + 4\sqrt{2} + 6\sqrt{6} = 10\sqrt{2} + 6\sqrt{6}$: C (the school's key).

33 Year 11 Advanced Exercise 5H Q7

Let $u = 2x$, so $0° \le u \le 720°$; solve for u, then halve.

a $u = 30°,\ 150°,\ 390°,\ 510°$: $x = 15°,\ 75°,\ 195°$ or $255°$

c $u = 135°,\ 225°,\ 495°,\ 585°$: $x = 67°30',\ 112°30',\ 247°30'$ or $292°30'$

d $u = 270°,\ 630°$: $x = 135°$ or $315°$

34 Year 11 Advanced Exercise 5H Q8

Let u be the compound angle; find its domain, solve for u, then find α.

a $u = \alpha - 45°$, $-45° \le u \le 315°$: $u = 30°,\ 210°$, $\alpha = 75°$ or $255°$

b $u = \alpha + 30°$, $30° \le u \le 390°$: $u = 240°,\ 300°$, $\alpha = 210°$ or $270°$

c $u = \alpha + 60°$, $60° \le u \le 420°$: $u = 360°$, $\alpha = 300°$

d $u = \alpha - 75°$, $-75° \le u \le 285°$: $u = 135°,\ 225°$, $\alpha = 210°$ or $300°$

35 Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q8(c)

Let $u = \alpha - 45°$. Since $0° \le \alpha \le 360°$, $-45° \le u \le 315°$. $\tan u = -\dfrac{1}{\sqrt{3}}$ is negative, so u is in quadrant 2 or 4, related angle 30°: $u = 150°$, $u = 330°$ (too big) or $u = -30°$. So $u = -30°$ or $150°$, and $\alpha = u + 45°$ gives $\alpha = 15°$ or $195°$.

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T2 | Exact values and trigonometric equations

Past-Paper Homework - Worked Solutions

PP 1 **Girraween High School, 2023, Year 10 Mathematics Yearly Exam, Short Answer Q9**

$\cos 30° = \sin(90° - 30°) = \sin 60°$ (complementary angles). So $\sin x = \sin 60°$ with $0 < x < 90°$, and $x = 60°$.

PP 2 **North Sydney Girls High School, 2024, Year 10 Mathematics Yearly Examination, Q10**

$\cos(90° - \theta) = \sin\theta$ (complementary angles) and $\sin(180° - \theta) = \sin\theta$ (Lesson T1). So the expression is $\dfrac{\sin\theta}{\sin\theta} = 1$. The answer is A.

PP 3 **Northern Beaches Christian School, 2023, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q7**

$\sin\theta$ is negative in quadrants 3 and 4, and the related angle is 30°. So $\theta = 180° + 30° = 210°$ or $360° - 30° = 330°$. The answer is A. Options B, C and D each contain an angle where $\sin\theta = +\frac{1}{2}$.

PP 4 **Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q16**

In the right-angled ΔAPT (T the top of the flagpole), h is opposite the 30° angle and $AP = 30$ is adjacent. $\tan 30° = \dfrac{h}{30}$, so $h = 30\tan 30° = \dfrac{30}{\sqrt{3}} = \dfrac{30\sqrt{3}}{3} = 10\sqrt{3}$.

The answer is B. Option C is $\dfrac{10}{\sqrt{3}}$, a third of this.

PP 5 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q16(d)**

150° is in quadrant 2, related angle 30°, and tan is negative there: $\tan 150° = -\tan 30° = -\dfrac{1}{\sqrt{3}}$. Also $\sin 45° = \dfrac{1}{\sqrt{2}}$.

$$\tan 150° + \sin 45° = -\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}} = \frac{-\sqrt{2} + \sqrt{3}}{\sqrt{6}}$$

$$= \frac{(\sqrt{3} - \sqrt{2})\sqrt{6}}{6} = \frac{\sqrt{18} - \sqrt{12}}{6} = \frac{3\sqrt{2} - 2\sqrt{3}}{6}$$

The exact value is $\dfrac{3\sqrt{2} - 2\sqrt{3}}{6}$.

PP 6 **Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q7(b)**

$\cos\theta$ is positive, so θ is in quadrant 1 or 4. Take $x = 2$, $r = \sqrt{5}$: $y^2 = 5 - 4 = 1$, so $y = \pm 1$. Hence $\sin\theta = \pm\dfrac{1}{\sqrt{5}}$ (both values are needed).

PP 7 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q17(b)**

$\sqrt{3}\tan\theta = 2 - 1 = 1$, so $\tan\theta = \dfrac{1}{\sqrt{3}}$. It is positive, so θ is in quadrant 1 or 3, related angle 30°: $\theta = 30°$ or $210°$.

PP 8 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q28**

$\sin(180° - \theta) = \sin\theta$ and $\sin(90° - \theta) = \cos\theta$, so the equation is $\sin\theta = \cos\theta$. Here $\cos\theta \neq 0$ (otherwise $\sin\theta = \pm 1 \neq 0$), so dividing by $\cos\theta$, $\tan\theta = 1$. Quadrant 1 or 3, related angle 45°: $\theta = 45°$ or $225°$.

PP 9 **Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Q25(a)**

(i) $\sin\theta$ positive: quadrant 1 or 2. Related angle $\sin^{-1} 0.416 \fallingdotseq 24.58°$. $\theta \fallingdotseq 24.58°$ or $180° - 24.58° = 155.42°$, so $\theta \fallingdotseq 25°$ or $155°$.
(ii) $\cos\theta$ negative: quadrant 2 or 3. Related angle $\cos^{-1} 0.302 \fallingdotseq 72.42°$.
$\theta \fallingdotseq 180° - 72.42° = 107.58°$ or $180° + 72.42° = 252.42°$, so $\theta \fallingdotseq 108°$ or $252°$.

PP 10 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part C, Q6**

$20\cos x° + 147 = 150$, so $20\cos x° = 3$ and $\cos x° = \dfrac{3}{20}$. Cos is positive, so x is in quadrant 1 or 4: $x = \cos^{-1}\dfrac{3}{20} \fallingdotseq 81.37$ or $x = 360 - 81.37 = 278.63$. The first two values are $x \fallingdotseq 81.37$ and 278.63 (about 81° and 279°).

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T3 | The ambiguous case

Lesson Examples

C1 **Tutor demo.** $\text{area} = \frac{1}{2} \times 5 \times 8 \times \sin 120°$. Since $\sin 120° = +\sin 60° = \dfrac{\sqrt{3}}{2}$ (quadrant 2), $\text{area} = 20 \times \dfrac{\sqrt{3}}{2} = 10\sqrt{3}$ square units.

Your turn. $\text{area} = \frac{1}{2} \times 6 \times 10 \times \sin 150° = 30 \times \frac{1}{2} = 15$ square units, since $\sin 150° = \sin 30° = \frac{1}{2}$.

C2 **Tutor demo.**

a $40 = \frac{1}{2} \times 15 \times x \times \sin 58°$, so $x = \dfrac{80}{15 \sin 58°} \div 6.289$ m.

b $30 = \frac{1}{2} \times 10 \times 9 \times \sin\theta = 45 \sin\theta$, so $\sin\theta = \frac{2}{3}$. Then $\theta \div 41.81°$ or $180° - 41.81°$, that is, $\theta \div 41°49'$ or $138°11'$.

Your turn.

a $50 = \frac{1}{2} \times 20 \times x \times \sin 64° = 10x \sin 64°$, so $x = \dfrac{5}{\sin 64°} \div 5.563$ m.

b $21 = \frac{1}{2} \times 8 \times 7 \times \sin\theta = 28 \sin\theta$, so $\sin\theta = \frac{3}{4}$, and $\theta \div 48°35'$ or $131°25'$.

C3 **Tutor demo.** $\angle A$ is opposite $BC = 5\sqrt{6}$ and $\angle B = 45°$ is opposite $AC = 10$.

$\dfrac{\sin A}{5\sqrt{6}} = \dfrac{\sin 45°}{10}$, so $\sin A = \dfrac{5\sqrt{6}}{10} \times \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{3}}{2}$, and $A = 60°$ or $120°$. Check: $60° + 45° = 105°$ and $120° + 45° = 165°$ are both less than 180°, so $A = 60°$ or $120°$.

Your turn. $\dfrac{\sin P}{4\sqrt{2}} = \dfrac{\sin 30°}{4}$, so $\sin P = \dfrac{4\sqrt{2}}{4} \times \dfrac{1}{2} = \dfrac{1}{\sqrt{2}}$, and $P = 45°$ or $135°$. Both are possible: $45° + 30° = 75°$ and $135° + 30° = 165°$. So $P = 45°$ or $135°$.

C4 **Tutor demo.** $\angle A$ is opposite $BC = 5$, and $\angle C = 75°$ is opposite $AB = 8$.

$\sin A = \dfrac{5 \sin 75°}{8} \div 0.6037$, so $A \div 37°8'$ or $142°52'$. But $142°52' + 75° > 180°$, so $A \div 37°8'$ is the only solution.

Your turn. $\sin X = \dfrac{6 \sin 70°}{9} \div 0.6265$, so $X \div 38°47'$ or $141°13'$. But $141°13' + 70° > 180°$, so $X \div 38°47'$ is the only solution.

Homework: Easy

1 Year 11 Advanced Exercise 5I Q1

a $x = \dfrac{10\sin 50°}{\sin 70°} \doteqdot 8.2$ **b** $x = \dfrac{3\sin 110°}{\sin 40°} \doteqdot 4.4$ **c** $x = \dfrac{4\sin 85°}{\sin 55°} \doteqdot 4.9$

d $x = \dfrac{2\sin 65°}{\sin 75°} \doteqdot 1.9$ **e** $x = \dfrac{7\sin 71°}{\sin 46°} \doteqdot 9.2$ **f** $x = \dfrac{5\sin 38°}{\sin 118°} \doteqdot 3.5$

In each part, x and the given side are both opposite given angles:

$$\frac{x}{\sin(\text{angle opposite } x)} = \frac{\text{given side}}{\sin(\text{its opposite angle})}.$$

2 Year 11 Advanced Exercise 5I Q2

a $a = \dfrac{12\sin 70°}{\sin 50°} \doteqdot 14.72$ **b** third angle 80°: $b = \dfrac{20\sin 80°}{\sin 25°} \doteqdot 46.61$

c third angle 20°: $c = \dfrac{15\sin 20°}{\sin 112°} \doteqdot 5.53$

3 Year 11 Advanced Exercise 5I Q3

In each part the known angle is opposite the longer given side, so only the acute answer is possible.

a $\sin\theta = \dfrac{4\sin 70°}{5}$: 49° **b** $\sin\theta = \dfrac{8\sin 85°}{10}$: 53°

c $\sin\theta = \dfrac{9\sin 100°}{13}$: 43° **d** $\sin\theta = \dfrac{6\sin 72°}{17}$: 20°

e $\sin\theta = \dfrac{7\sin 124°}{12}$: 29° **f** $\sin\theta = \dfrac{16\sin 53°}{19}$: 42°

4 Year 11 Advanced Exercise 5I Q4

a $\frac{1}{2} \times 3 \times 4 \times \sin 50° \doteqdot 4.60$: 5 cm²

b $\frac{1}{2} \times 5 \times 8 \times \sin 70° \doteqdot 18.79$: 19 cm²

c $\frac{1}{2} \times 7 \times 7 \times \sin 115° \doteqdot 22.20$: 22 cm²

5 Northern Beaches Christian School, 2022, Mathematics Year 10 5.3/Year 9 Accelerated Yearly Examination, Section I Q5

The 35° angle is between the sides 30 m and 20 m.

$\text{area} = \frac{1}{2} \times 30 \times 20 \times \sin 35° \doteqdot 172.07$, so the area is about 172 m². The answer is C. The side 17.8 m is not needed.

6 Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q20

a The side b is opposite 61° and 8 cm is opposite 42°. $b = \dfrac{8\sin 61°}{\sin 42°} \doteqdot 10.46$, so $b \doteqdot 10$ (2 s.f.).

b The third angle is $180° - 61° - 42° = 77°$, between the sides 8 and b.

$\text{area} = \frac{1}{2} \times 8 \times 10.46 \times \sin 77° \doteqdot 40.76$, so the area is about 41 cm². (Use the unrounded value of b.)

7 Girraween High School, 2022, Year 10 Yearly Examination Mathematics, Q25(b)

θ is opposite 416.6, and $74°24'$ is opposite 515.5.

$\sin\theta = \dfrac{416.6\sin 74°24'}{515.5} \doteqdot 0.7784$, so $\theta \doteqdot 51°7'$ or $128°53'$. The second is impossible, because $128°53' + 74°24' > 180°$. So $\theta \doteqdot 51°7'$.

Homework: Medium

8 Year 11 Advanced Exercise 5I Q5

b $C = 180° - 43° - 101° = 36°$. $b = \dfrac{7.5\sin 101°}{\sin 43°} \doteqdot 10.80$ cm,

$c = \dfrac{7.5\sin 36°}{\sin 43°} \doteqdot 6.46$ cm.

9 Year 11 Advanced Exercise 5I Q6

b $X = 180° - 58° - 52° = 70°$. $x = \dfrac{32\sin 70°}{\sin 58°} \doteqdot 35.46$,

$z = \dfrac{32\sin 52°}{\sin 58°} \doteqdot 29.73$. Perimeter $\doteqdot 32 + 35.46 + 29.73 = 97.19$, about 97 cm.

10 Year 11 Advanced Exercise 5I Q7

a $\sin B = \dfrac{2.7\sin 52°21'}{2.8} \doteqdot 0.7636$, so $B \doteqdot 49°46'$ or $130°14'$. The second is impossible, because $130°14' + 52°21' > 180°$ (also $b < a$, so $B < A$). So $B \doteqdot 49°46'$.

b $C = 180° - 52°21' - 49°46' = 77°53'$.

c $\text{area} = \frac{1}{2} \times 2.8 \times 2.7 \times \sin 77°53' \doteqdot 3.70$ cm².

11 Year 11 Advanced Exercise 5I Q8

$\dfrac{\sin\theta}{9} = \dfrac{\sin 22°}{5}$, so $\sin\theta = \dfrac{9\sin 22°}{5} \doteqdot 0.6743$, and $\theta \doteqdot 42°$ or $138°$. Both are possible: $138° + 22° = 160°$.

12 Year 11 Advanced Exercise 5I Q9

$\sin\theta = \dfrac{6\sin 36°}{4} \doteqdot 0.8817$, so $\theta \doteqdot 62°$ or $118°$. Both are possible: $118° + 36° = 154°$.

13 Year 11 Advanced Exercise 5I Q10

a $\sin Q = \dfrac{15\sin 25°50'}{7} \doteqdot 0.9338$, so $Q \doteqdot 69°2'$ or $110°58'$ (both possible, since $110°58' + 25°50' < 180°$).

b If $Q \doteqdot 69°2'$, then $R \doteqdot 85°8'$ and $r = \dfrac{7\sin R}{\sin 25°50'} \doteqdot 16.0$ cm. If $Q \doteqdot 110°58'$, then $R \doteqdot 43°12'$ and $r \doteqdot 11.0$ cm.

14 Year 11 Advanced Exercise 5I Q11

In ΔABC: $\angle A = 39°$, $\angle ABC = 180° - 63° = 117°$ (straight angle along the north line through B), so $\angle C = 24°$. $AC = \dfrac{67 \sin 117°}{\sin 24°} \doteqdot 146.77$ and $BC = \dfrac{67 \sin 39°}{\sin 24°} \doteqdot 103.67$. Distance $\doteqdot 67 + 146.77 + 103.67 = 317.44$, about 317 km.

15 Year 11 Advanced Exercise 5I Q12

a The elevation 34° and the path's 12° are both measured from the horizontal at A, so $\angle TAB = 34° - 12° = 22°$. At B, the vertical pole makes 90° with the horizontal, and the path below B is a further 12° below the horizontal, so $\angle ABT = 90° + 12° = 102°$.

b $\angle ATB = 180° - 22° - 102° = 56°$. $TB = \dfrac{20 \sin 22°}{\sin 56°} \doteqdot 9.04$, so the pole is about 9 m high.

16 Year 11 Advanced Exercise 5I Q13

a $\dfrac{b}{\sin B} = \dfrac{a}{\sin A}$, so $b = \dfrac{12 \times \frac{2}{3}}{\frac{1}{4}} = 32$. **b** $\sin P = \dfrac{p \sin Q}{q} = \dfrac{25 \times \frac{3}{5}}{21} = \dfrac{15}{21} = \dfrac{5}{7}$

17 Year 11 Advanced Exercise 5I Q14

a $48 = \frac{1}{2} \times 12 \times x \times \sin 30° = 3x$: $x = 16$ m **b** $x = \dfrac{96}{9 \sin 70°} \doteqdot 11.35$ m

c $x = \dfrac{96}{36 \sin 130°} \doteqdot 3.48$ m

18 Year 11 Advanced Exercise 5I Q15

a $72 = \frac{1}{2} \times 16 \times 18 \sin\theta = 144 \sin\theta$, $\sin\theta = \frac{1}{2}$: $\theta = 30°$ or $150°$

b $72 = 240 \sin\theta$, $\sin\theta = 0.3$: $\theta \doteqdot 17°27'$ or $162°33'$

c $72 = 60 \sin\theta$, $\sin\theta = 1.2$: no solutions, because $\sin\theta$ cannot be greater than 1.

19 North Sydney Boys High School, 2022, Stage 5.3 Mathematics Year 10 Assessment Task 4, Section V Q10

a) A sketch with $\angle B = 60°$, $AB = c = 24$ (opposite C) and $AC = b = 31$ (opposite B).

b) $\dfrac{\sin C}{24} = \dfrac{\sin 60°}{31}$, so $\sin C = \dfrac{24 \sin 60°}{31} \doteqdot 0.6705$, and $C \doteqdot 42°6'$ or $137°54'$. But $137°54' + 60° > 180°$, so $\angle ACB \doteqdot 42°6'$.

Homework: Hard

20 Year 11 Advanced Exercise 5I Q16

a $x = \dfrac{6\sin 60°}{\sin 45°} = 6 \times \dfrac{\sqrt{3}}{2} \times \sqrt{2} = 3\sqrt{6}$

b $x = \dfrac{6\sin 30°}{\sin 45°} = 6 \times \frac{1}{2} \times \sqrt{2} = 3\sqrt{2}$

c $x = \dfrac{6\sin 45°}{\sin 60°} = \dfrac{6}{\sqrt{2}} \times \dfrac{2}{\sqrt{3}} = \dfrac{12}{\sqrt{6}} = 2\sqrt{6}$

d $x = \dfrac{6\sin 45°}{\sin 30°} = \dfrac{6}{\sqrt{2}} \times 2 = 6\sqrt{2}$

21 Year 11 Advanced Exercise 5I Q17

Let each equal side be s cm. $35 = \frac{1}{2}s^2 \sin 35°$, so $s^2 = \dfrac{70}{\sin 35°} \div 122.04$ and $s \div 11.05$, about 11.0 cm.

22 Year 11 Advanced Exercise 5J Q14

$7^2 = x^2 + 8^2 - 2 \times x \times 8 \times \cos 60°$, so $49 = x^2 + 64 - 8x$, $x^2 - 8x + 15 = 0$, $(x-3)(x-5) = 0$, and $x = 3$ or 5.

23 Year 11 Advanced Exercise 5J Q15

a $4^2 = x^2 + 6^2 - 2 \times x \times 6 \times \cos 30° = x^2 + 36 - 6\sqrt{3}\,x$, so $x^2 - 6\sqrt{3}\,x + 20 = 0$.

b $x = \dfrac{6\sqrt{3} \pm \sqrt{108 - 80}}{2} = \dfrac{6\sqrt{3} \pm 2\sqrt{7}}{2} = 3\sqrt{3} \pm \sqrt{7}$.

c The other position of C is on AC, nearer to A, at distance $3\sqrt{3} - \sqrt{7} \div 2.55$ cm from A, so that $BC = 4$ cm again (the two positions are reflections of each other in the perpendicular from B to line AC).

24 Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, second Q4

$\angle ACD = 180° - 130° = 50°$ (straight angle). In ΔACD, $\angle ADC$ is opposite $AC = 7.9$ and 50° is opposite $AD = 6.4$: $\sin \angle ADC = \dfrac{7.9 \sin 50°}{6.4} \div 0.9456$, so $\angle ADC \div 71°$ or $109°$. The acute angle is $\angle ADC \div 71°$.

25 Fort Street High School, 2022, Year 10 Common Task 2, Mathematics, Question 6(b)

θ is opposite 11.7, and $46°$ is opposite 8.9. $\dfrac{\sin\theta}{11.7} = \dfrac{\sin 46°}{8.9}$, so $\sin\theta = \dfrac{11.7 \sin 46°}{8.9} \div 0.9457$, and $\theta \div 71.02°$ or $180° - 71.02° = 108.98°$. Both are possible: $46° + 71° < 180°$ and $46° + 109° < 180°$. So $\theta \div 71°$ or $109°$ (the school's answer).

26 Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(c)

i. Since Q is due east of P, $\angle RPQ = 90° - 70° = 20°$. By the cosine rule, $RQ^2 = 12^2 + 30^2 - 2 \times 12 \times 30 \times \cos 20° \doteqdot 367.43$, so $RQ \doteqdot 19.17$, about 19 km.

ii. $\sin \angle PQR = \dfrac{12 \sin 20°}{19.17} \doteqdot 0.2141$, so $\angle PQR \doteqdot 12.36°$ (acute, because it is opposite the shortest side). From Q, due west is 270°T and R is $12.36°$ north of west, so the bearing of R from Q is $270° + 12.36° \doteqdot$ 282°T.

Year 10 Trigonometry (part 2) - Homework Answers

Lesson T3 | The ambiguous case

Past-Paper Homework - Worked Solutions

PP 1 **Hornsby Girls High School, 2021, Year 10 Mathematics Term 4 Assessment, Q9**

$\angle F$ is opposite $f = 7$, and $\angle D = 42°$ is opposite $d = 5$. $\sin F = \dfrac{7 \sin 42°}{5} \doteqdot 0.9368$, so $F \doteqdot 69.52°$ or $180° - 69.52° = 110.48°$. The obtuse value is possible, since $110.48° + 42° < 180°$, and to the nearest degree it is 110°. The answer is C. (B is the acute value; A and D do not have the right sine.)

PP 2 **The Scots College, 2023, Year 10 Assessment Task 4, Mathematics 5.3, Q19**

θ is opposite 12.8 cm and $21°23'$ is opposite 5.9 cm. $\dfrac{\sin\theta}{12.8} = \dfrac{\sin 21°23'}{5.9}$, so $\sin\theta \doteqdot 0.7911$ and the acute angle is $52.28°$. Since θ is obtuse, $\theta = 180° - 52.28° = 127.72°$, that is, $\theta \doteqdot 127°43'$.

PP 3 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, (b) (i)**

$\angle C$ is opposite $AB = 5$ and $\angle B = 25°$ is opposite $AC = 3$.
$\sin C = \dfrac{5 \sin 25°}{3} \doteqdot 0.7044$, so the acute value is about 44.78°. $\angle C$ is obtuse, so $\angle C = 180° - 44.78° \doteqdot 135.22°$, about 135°.

PP 4 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q14**

θ is opposite the side 10, and 35° is opposite the side 6. $\dfrac{\sin\theta}{10} = \dfrac{\sin 35°}{6}$, so $\sin\theta \doteqdot 0.9560$ and $\theta \doteqdot 72.93°$ or $107.07°$. Both are possible: $107° + 35° = 142° < 180°$. So $\theta \doteqdot 73°$ or $107°$.

PP 5 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(b)**

i. The 60° angle is between the sides 8 cm and x cm. $\frac{1}{2} \times 8 \times x \times \sin 60° = 22\sqrt{3}$, so $4x \times \dfrac{\sqrt{3}}{2} = 22\sqrt{3}$, $2\sqrt{3}\,x = 22\sqrt{3}$, and $x = 11$.
ii. y is opposite the 60° angle.
$y^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 60° = 64 + 121 - 88 = 97$, so $y = \sqrt{97}$ cm.

PP 6 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part B, Q4**

In the right-angled ΔABC, $\sin A = \dfrac{BC}{AB} = \dfrac{8}{18} = \dfrac{4}{9}$. Also $AD = 18 + 6 = 24$. In ΔADE, the angle A is between AD and AE: $160 = \frac{1}{2} \times 24 \times AE \times \dfrac{4}{9} = \dfrac{16}{3}AE$, so $AE = \dfrac{480}{16} = 30$, that is, $AE = 30.00$ cm.

PP 7 **North Sydney Girls High School, 2023, Year 10 Mathematics Yearly Examination, Q20(a)**

$\angle BAC = 80° - 50° = 30°$. In ΔABC, $\angle ACB$ is opposite $AB = 150$ and 30° is opposite $BC = 80$: $\sin \angle ACB = \dfrac{150 \sin 30°}{80} = 0.9375$, so $\angle ACB \doteqdot 69°38'$ or $110°22'$, giving $\angle ABC \doteqdot 80°22'$ or $39°38'$. Since $AC > AB$, the angle opposite AC is larger than the angle opposite AB: $\angle ABC > \angle ACB$. Only $\angle ABC \doteqdot 80°22'$ (with $\angle ACB \doteqdot 69°38'$) satisfies this. She was heading along AB and turned towards C, so she turned through $180° - 80°22' = 99°38'$.

Year 10 Trigonometry (part 2) - Homework Answers

Chapter Review | Lessons T1 to T3

R 1 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Section A Q5**

The gradient is −1, so $\tan\alpha = -1$ with α obtuse: $\alpha = 180° - 45° = 135°$.

R 2 **North Sydney Girls High School, 2022, Year 10 Mathematics Yearly Examination, Q18(a)**

$\cos 45° = \dfrac{1}{\sqrt{2}}$, and $\tan 120° = -\tan 60° = -\sqrt{3}$ (quadrant 2). So the value is $\dfrac{1}{\sqrt{2}} - \sqrt{3}$.

R 3 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q22(a)**

$\tan\theta$ is negative in quadrants 2 and 4, related angle 45°: $\theta = 135°$ or $315°$.

R 4 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part D, first Q4**

$\cos 225° = -\cos 45° = -\dfrac{1}{\sqrt{2}}$, so $\cos x = \dfrac{1}{\sqrt{2}}$. Quadrants 1 and 4, related angle 45°: $x = 45°$ or $315°$.

R 5 **Fort Street High School, 2021, Year 10 Common Task 2, Mathematics, Question 6**

θ is opposite 25cm, and $20°$ is opposite 10cm. $\sin\theta = \dfrac{25\sin 20°}{10} \doteqdot 0.8551$, so $\theta \doteqdot 58.77°$ or $121.23°$. Both are possible ($20° + 121° < 180°$), so $\theta \doteqdot 59°$ or $121°$ (the school's answer).

R 6 **Sydney Grammar School, 2020, Annual Examination Form IV Mathematics, Q3(b)**

(i) $AB^2 = 5^2 + 6^2 - 2 \times 5 \times 6 \times \cos 60° = 25 + 36 - 30 = 31$, so $AB = \sqrt{31}$ cm.

(ii) $\text{area} = \frac{1}{2} \times 5 \times 6 \times \sin 60° = 15 \times \dfrac{\sqrt{3}}{2} = \dfrac{15\sqrt{3}}{2}$ cm².